

1. Create a table listing 3 types of printers, 2 types of scanners, and 2 types of webcams, and for each, write one real-life use case.

Device	Type	Real-life use case
Printer	Inkjet Printer	Printing college assignments and colour images
Printer	Laser Printer	Printing large numbers of documents in an office
Printer	3D Printer	Creating 3D models and prototypes
Scanner	Flatbed Scanner	Scanning certificates and documents
Scanner	Sheet-fed Scanner	Scanning multiple pages quickly in an office
Webcam	USB Webcam	Attending online classes or meetings
Webcam	Built-in Webcam	Video calling from a laptop

2. Draw a labeled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).

1. USB Connection



Laptop / PC

USB Cable
(Type A to Type B or Type A to C)



Printer

Interface: USB

A physical cable connects the laptop/PC to the printer for data transfer.

2. Wi-Fi Connection



Laptop / PC



Wi-Fi Router
(Home Network)



Printer

Interface: Wi-Fi

Both devices connect to the same wireless network (no cable required).

3. Bluetooth Connection



Laptop / PC



Bluetooth
(short-range wireless)



Printer

Interface: Bluetooth

Uses short-range wireless technology to connect the laptop/PC to the printer.

- **USB:** A wired connection used to connect devices directly to a computer. It is fast and reliable but needs a cable.
- **Wi-Fi:** A wireless connection that allows devices to communicate through a network. It is convenient but requires network setup.
- **Bluetooth:** A short-range wireless connection between compatible devices. It is easy to use but has limited range.

3. List three different connection interfaces (such as USB, Wi-Fi, Bluetooth) used by peripherals like printers, scanners, and webcams, and for each, name one advantage and one disadvantage.

Connection Interface	Advantages	Disadvantages
USB	• Fast data transfer. • Reliable and stable connection. • Easy to install and use. • Widely compatible with computers. • Does not require an internet connection.	• Requires a physical cable. • Limited cable length. • One device may need to be connected directly to the computer. • Cables can create clutter. • USB ports may be limited on some laptops.
Wi-Fi	• No physical cable is required. • Multiple computers can share one peripheral. • Convenient for home and office use. • Allows devices to be placed farther from the computer. • Easy to use with laptops and mobile devices.	• Requires Wi-Fi/network setup. • Connection may be affected by weak signals. • Can be slower than a direct USB connection. • Network problems can interrupt communication. • Some devices may require additional configuration.
Bluetooth	• Wireless and cable-free. • Easy to pair compatible devices. • Low power consumption. • Useful for short-distance connections. • Does not require a Wi-Fi router.	• Limited operating range. • Usually slower than USB. • Both devices must support Bluetooth. • Connection can sometimes be interrupted. • Not ideal for transferring very large amounts of data.

4. Pick any one device you use at home or college (printer, scanner, or webcam). Write step-by-step instructions for how you would connect it to your computer, including what cable or wireless method you would use and any setup steps required.

Device: HP LaserJet M1005

Connection Method: USB Cable

- **Step 1:** Place the HP LaserJet M1005 on a stable, flat surface near the computer.
- **Step 2:** Connect the printer's power cable to the power supply and switch on the printer.
- **Step 3:** Take the USB printer cable and prepare it for connection.
- **Step 4:** Connect one end of the USB cable to the printer.
- **Step 5:** Connect the other end of the USB cable to an available USB port on the computer
- **Step 6:** Wait for the computer to detect the printer and install the required driver.
- **Step 7:** Open Settings → Printers & Scanners on the computer.
- **Step 8:** Check that HP LaserJet M1005 appears in the list of printers.
- **Step 9:** Select the printer and set it as the default printer if required.
- **Step 10:** Open a document and select the Print option.
- **Step 11:** Select HP LaserJet M1005 and print a test page.
- **Step 12:** Check the printed page to confirm that the printer is working correctly.
- **Result:** The HP LaserJet M1005 was successfully connected to the computer using a USB cable.